

Time Dilation and Length Contraction



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Whatever we will discuss

- Time dilation and its explanation
- Deduction of the equation, $t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$
- Length contraction and its explanation
- Deduction of the equation, $L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$
- Mathematical problems related to Time dilation, Length contraction and Relativity of Mass

Time Dilation

Time Dilation

- The time interval between the occurrence of two events which are in motion with respect to an observer will be longer than the time interval of occurrence of the same two events which are at rest with respect to the observer. This is called time dilation.

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

If $v = 0.98c$ then $t = 5t_0$

If $v = 0.999c$ then $t = 22.3t_0$

Deduction of Time Dilation Equation

$$\begin{aligned}t &= t_2 - t_1 \\&= \frac{t_2' + vx' / c^2}{\sqrt{1 - v^2 / c^2}} - \frac{t_1' + vx' / c^2}{\sqrt{1 - v^2 / c^2}} \\&= \frac{t_2' - t_1'}{\sqrt{1 - v^2 / c^2}} \\t &= \frac{t_0}{\sqrt{1 - v^2 / c^2}}\end{aligned}$$

Length Contraction

Length Contraction

- The length of an object in motion with respect to an observer appears to be shorter than when it is at rest with respect to him, this phenomenon is known as length contraction.

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

Deduction of Length Contraction Equation

$$\begin{aligned} L_0 &= x_2 - x_1 \\ &= \frac{x_2' + vt'}{\sqrt{1 - v^2/c^2}} - \frac{x_1' + vt'}{\sqrt{1 - v^2/c^2}} \\ &= \frac{x_2' - x_1'}{\sqrt{1 - v^2/c^2}} = \frac{L}{\sqrt{1 - v^2/c^2}} \\ L &= L_0 \sqrt{1 - v^2/c^2} \end{aligned}$$

Related Mathematical Problems

❖ An astronaut at the age of 30 years travels in a spaceship with a velocity of $2.4 \times 10^8 \text{ ms}^{-1}$ and returns after 50 years. What is his present age?
[Ans. 60 year]

❖ The length of a rocket is measured half of the actual length. Calculate the speed of the rocket.
[Ans. $2.598 \times 10^8 \text{ ms}^{-1}$]

❖ The length of imaginary train is measured one third of the actual length. Calculate the speed of the train.
[Ans. $2.83 \times 10^8 \text{ ms}^{-1}$]

❖ The length of a rocket on the ground is 100 m. When rocket moves with respect to a stationary observer on the ground its length appears to be 99 m. Calculate the speed of the rocket.
[Ans. $4.3 \times 10^7 \text{ ms}^{-1}$]

❖ An electron is moving with a speed of $0.99c$, what is its moving mass? Rest mass of electron is $9.1 \times 10^{-31} \text{ kg}$.
[Ans. $64.5 \times 10^{-31} \text{ kg}$]

❖ Mass of 40 years old Abid is 55 kg. He went to investigate the moon by a spaceship moving with a velocity of $2.4 \times 10^8 \text{ ms}^{-1}$. When the age of his twin brother Rayan is 80 year he returned to the earth. (a) Find the mass of Abid in the spaceship. (b) What is his present age of Abid?

[Ans. (a) 91.67 kg and (b) 64 year]

❖ A 10 m long object of mass 20 kg starts moving from rest with a velocity $0.5c$. What is the length of the object when in motion?

[Ans. 8.66 m]

❖ The length and mass of a rocket on earth are 10 m and 5000 kg respectively. It started moving with a velocity $3 \times 10^7 \text{ ms}^{-1}$ with respect to a stationary observer on earth. (a) Find the moving length of the rocket. (b) If the velocity of the rocket is doubled, what change in its mass will take place?

[Ans. (a) 9.95 m and (b) the mass will be increased by 77.915 kg]



Thank You
Thanks for your attention